
Archaeological Survey in Eastern Guadalajara Province, Spain: Initial Results

A. Burke[†], J. Maíllo Fernández[‡], N. Fuertes Prieto[§], M. Bisson[£],
P. James[¥], and G. Levavasseur[¶]

ABSTRACT. The Palaeolithic record of the interior of the Iberian Peninsula, the central Meseta, is still relatively poorly documented. In this article we report the results of a preliminary, stratified survey conducted in the eastern part of the province of Guadalajara during 2009 and 2010. The survey enabled us to identify a series of Middle and Upper Palaeolithic sites in an area where no previously recorded Palaeolithic sites exist. These results, in turn, are used to propose a predictive model of archaeological potential that will guide further surveys in the target region.

RÉSUMÉ. L'histoire de l'occupation paléolithique de l'intérieur de la Péninsule ibérique, la Meseta centrale, est encore méconnue. Une prospection archéologique menée de 2009 à 2010 dans la zone Est de la province de Guadalajara nous a permis d'identifier une série inédite de sites du Paléolithique moyen et supérieur dont nous présentons les plus significatifs ici. Ces résultats nous permettent également de formuler un modèle prédictif de potentiel archéologique qui guidera les prospections futures dans cette région.

OUR KNOWLEDGE OF THE PALAEOlithic record of the central Meseta, the highland plateau located in the interior of the Iberian Peninsula (Figure 1), is uneven both in spatial and chronological terms. The contrast between the rich archaeological record of coastal Iberia, and the much sparser

record of the interior, has led to the suggestion that the central Meseta suffered periodic population declines during prehistory, particularly during the Middle and Upper Palaeolithic. The continentality of the region, its relative elevation, and mountainous character (the Cantabrian Cordillera bordered by the Sierra Morena and Sistema Ibérico with the Sistema Central at its heart) are sometimes evoked as factors affecting the distribution of human populations, particularly during colder periods of the Late Pleistocene.

[†]Corresponding author:
Université de Montréal, Anthropologie,
C.P. 6128 Centre-Ville, Montréal, QC,
Canada H3T 3J7 [a.burke@umontreal.ca]

[‡]Departamento de Prehistoria y Arqueología
de la Universidad Nacional de Educación a
Distancia, C/Senda del Rey 7 28040 MADRID
Madrid, Spain [jlmaillo@geo.uned.es]

[§]Área de Prehistoria, Facultad de Filosofía
y Letras, Universidad de León, Campus de
Vegazana, Spain [mnfuep@unileon.es]

[£]McGill University, Department of Anthropology,
Leacock Building, 855 Sherbrooke Street West
Montreal, QC, Canada H3A 2T7
[michael.bisson@mcgill.ca]

[¥]Université de Montréal, Sciences biologiques,
bureau F056, Pavillon Marie-Victorin, Montreal,
QC, Canada [patrick.ma.james@umontreal.ca]

[¶]Laboratoire des Sciences du Climat et de
l'Environnement, LSCE/IPSL - CEA-CNRS-
UVSQ, CE Saclay, l'Orme des Merisiers,
batiment 701 91191 Gif-sur-Yvette Cedex France
[guillaume.levavasseur@lsce.ipsl.fr]

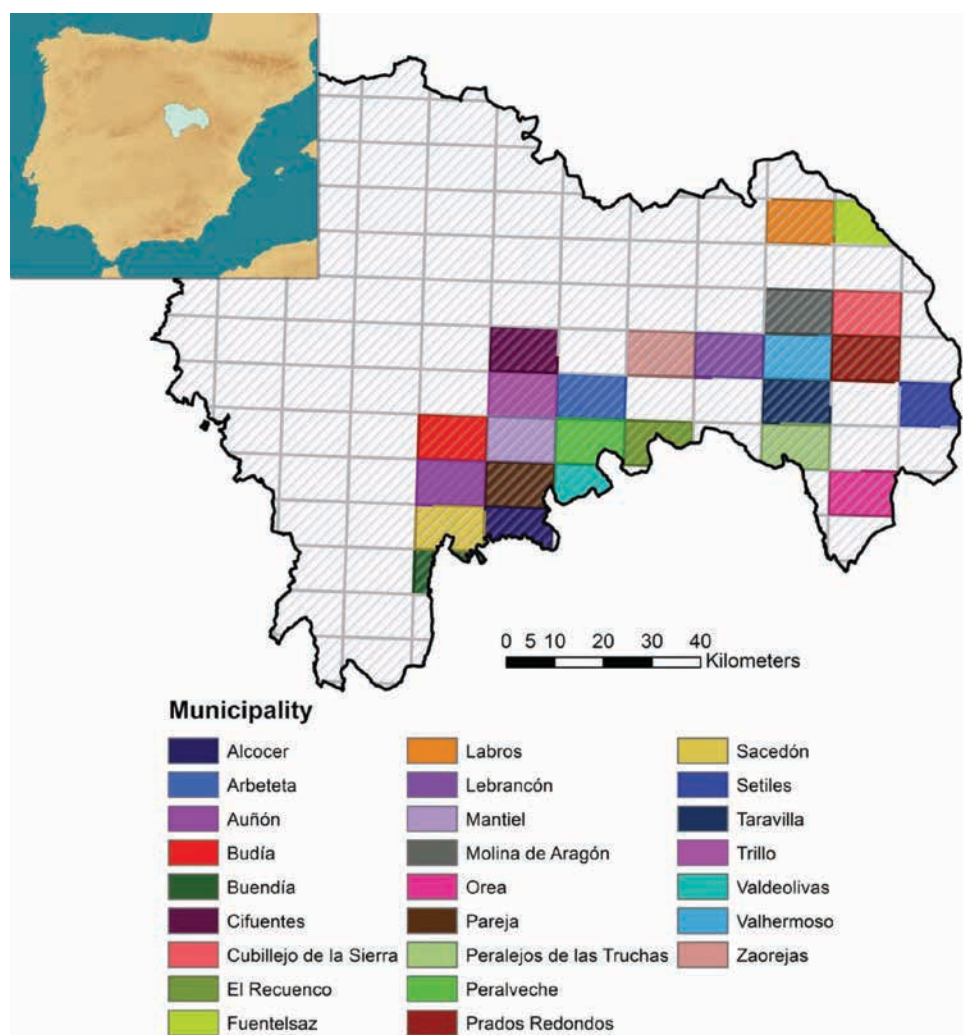


FIGURE 1. Survey region (eastern Guadalajara province) showing municipalities where surveys were undertaken (2009, 2010). Inset: Location of Guadalajara within the Iberian Peninsula.

Recent fieldwork has added significantly to the existing archaeological record of the central Meseta, altering our perception of population dynamics during the Palaeolithic period in the region. The distribution of research effort in the central Meseta remains somewhat uneven, however, with some regions—such as the province of Burgos—receiving more attention than

others. This is due to significant discoveries made at Atapuerca (at Gran Dolina and Sima de los Huesos) during the late twentieth century which, among other things, provided a key to our current understanding of the initial Lower Palaeolithic occupation of the European continent (Aguirre 2001; Carbonell et al. 1995; López-García et al. 2010; Mosquera et al. 2013). Other important dis-

coveries in the Central Meseta include well-known sites such as Ambrona and Torralba in Soría, Áridos in Madrid, and Pinedo in Toledo (Santonja and Pérez González 2000/2001, 2005).

The Middle Palaeolithic in the central Meseta is documented by surface collections as well as stratified sites such as Atapuerca (Rodríguez 2004), Prado Vargas (Navazo and Díez 2008; Navazo et al. 2005), Cueva Millán and La Ermita (Moure Romanillo and García-Soto Mateos 1983), Cueva Corazón (Sánchez et al. 2011), El Cañaveral (Baena et al. 2008), Navalmaíllo, Pinilla del Valle, and Buena Pinta (Arsuaga et al. 2011; Navazo and Díez 2008; Navazo et al. 2005), as well as sites in the Manzanares and Jarama River Basins, in the province of Madrid (Panera Gallego 2010; Rubio 2011).

The Upper Palaeolithic is relatively poorly represented in the central Meseta. Known sites include: the early Upper Palaeolithic site of Valle de las Orquídeas (Mosquera et al. 2007); El Palomar (Vega Toscano and Martín Blanco 2006); the Magdalenian site of Peña de Estebanvela, Vergara; El Monte, near Madrid (Cacho et al. 2010; Vega et al. 2010); and Verdelpino and Buendia, in Cuenca (de la Torre et al. 2007; Rasilla Vives et al. 1996). On the eastern border of the Meseta, evidence for Upper Palaeolithic occupations of the *Sistema Ibérico* has also recently been reported (Utrilla et al. 2010). In addition to these stratified archaeological sites, sites with Upper Palaeolithic art are known in the northern part of the Meseta, including Domingo García (Ripoll and Municio 1999), La Griega in Segovia (Corchón 1997) and placa de Villalba, Soria (Jimeno Martínez et al. 1990).

Where research effort has been extended in the central Meseta, new

discoveries have filled some of the gaps in our knowledge of the Palaeolithic settlement of the region. It seems plausible, therefore, to suggest that biases in the archaeological record of the central Meseta exist and should be corrected. In order to act on this suggestion, we developed a survey project focussing on the province of Guadalajara, within the central Meseta, which has a relatively poorly documented history of Palaeolithic occupation.

The Study Region: Guadalajara Province

The province of Guadalajara is located south of Madrid in the very center of the Meseta (Figure 1). Archaeological sites dated to the Palaeolithic are relatively scarce and unevenly distributed in this region (Malla 1997) and research has mostly focussed on the Jarama and Sorbe River Basins, where a series of sites with relatively short stratigraphic sequences have been discovered (García et al. 1992). Lower Palaeolithic sites are rare in Guadalajara despite their relative abundance in the nearby province of Madrid. Sites such as Jarama I and VI (Alcolea González et al. 1998; García Valero 1997; García et al. 1992), Peña Capón and Torrejones (Arribas et al. 1995), and Cueva de los Casares (Barandiarán 1973; Beltrán and Barandiarán 1968) bear witness to the Middle Palaeolithic occupation of the province. Solutrean and Proto-Solutrean occupations at Peña Capón are the earliest known Upper Palaeolithic occupations, followed by later Upper Palaeolithic sites such as Jarama II, El Turismo, Cueva del Reno, and los Enebrales (de la Torre 2007). Examples of Upper Palaeolithic parietal art in Guadalajara occur at Cueva de los Casares (Barandiarán 1973), Cueva de la Hoz (Balbín Berhmann et al. 1995;

Beltrán and Barandiarán 1968), El Reno and El Turismo (Alcolea González and de Balbín Behrmann 2003; Balbín Behrmann et al. 1995; Chapa 2006) and mobiliary art is present at Jarama II (Jordá Pardo and García Valero 1989).

When we began this project in 2009, however, few Palaeolithic sites were known in the eastern half of Guadalajara Province, along the upper reaches of the Tagus River and its tributaries, the exception being a Palaeolithic site (referred to as a “necropolis”) at the place named “La Olmedilla,” on the slopes above the Entrepénas reservoir, near Sacedón (Mercado Blanco et al. 2003). We therefore chose to concentrate our surveys in eastern Guadalajara.

The topography in the western part of the study region around the Entrepénas reservoir and the village of Sacedón is relatively gentle; to the east near Molina de Aragón the terrain is more rugged and mountainous. Rockshelters and caves are rare in the west due to the nature and composition of the local bedrock, though limestone formations occasionally outcrop sufficiently for small rockshelters and caves to form (e.g., more centrally, near Arbeteta and Trillo). Bedrock in the target region alternates between limestone and sandstone, both of which are deeply incised by river canyons, particularly in the east where the Tagus and Hoz Seca river valleys offer good potential for the formation of caves and rockshelters. Flint nodules outcrop fairly frequently in the western portion of the target region but are conspicuously rare in the east.

Methods

Field Surveys

The ultimate goal of this research is to document the history of Palaeolithic

occupation of eastern Guadalajara, a hitherto relatively unknown part of the Central Meseta. To this end, we conducted preliminary field surveys in 2009–2010 to assess the archaeological potential of the study region and establish a chronological framework. Results of the surveys (presence/absence of archaeological material) are used to produce an initial predictive model intended as a guide for the design of future fieldwork.

The surveys were carried out over the course of two one-month long field seasons during the summers of 2009 and 2010, with the aid of funding from the Social Sciences and Humanities Research Council of Canada (grant #401-2009-2240). The initial strategy for the field surveys was designed with the aid of geological (IGEO Magma 1:50,000 series) and topographical (CNIG MTN 1:25,000 series) maps, in addition to a speleological catalogue (Fernández Tabera and Martín Yebra 1982). We systematically targeted locations where limestone outcroppings and ancient river terraces are indicated on the geological maps and randomly surveyed the interfluvies, targeting bluffs and exploring any ploughed fields or recently turned soil encountered. We also targeted caves, using the speleological catalogue as a guide.

A total of 198 locations in a variety of environmental settings were surveyed by teams of two to four archaeologists (Figure 1). Survey tracks were systematically recorded and all lithic artefacts encountered were geo-located using hand-held GPS devices and collected. A preliminary study of a selection of the lithic remains was conducted by Bisson in the summer of 2010 in order to establish a chronological framework and assess the potential of the study

region in terms of the preservation of archaeological sites; the results of this study are reported below. Further lithic analyses are still underway (by J. Maíllo Fernández and N. Fuertes Prieto) at the Universidad de León and the Universidad Nacional de Educación a Distancia (Madrid).

Spatial Modelling

Spatial modelling of the occurrence of archaeological remains has been in use in archaeology since the 1960s, particularly in the field of Cultural Resource Management where models have been used predictively for mitigation and survey design (e.g., Dalla Bona 1994). Several excellent reviews of the historical application of spatial modelling in archaeology are available (e.g., Ebert 2004; Kvamme 2006; Wheatley and Gillings 2004). Spatial models tend to focus on environmental variables, partly due to their availability and inherent “mapability,” though this is less and less the case (McCoy and Ladefoged 2009). This focus has led to accusations of environmental determinism (Gaffney and van Leusen 1995). Environmental variables (such as elevation, slope, aspect, and soil type) are often good predictors of the likelihood of encountering archaeological remains, however, and have been shown to have explanatory potential when examining prehistoric human settlement strategies.

The predictors used in constructing spatial models in archaeology vary according to the type of model. Models constructed as tools for assessing archaeological potential (predictive models) do not need to be complicated in order to be effective (in fact, the reverse is true) and the use of simple environmental predictors and modern hydrology is perfectly justified. More sophisticated

interpretive (or explanatory) models, on the other hand, often incorporate variables that reflect the dynamic processes shaping the landscape as well as human behaviour (Bevan and Conolly 2002). Our goal in this initial research is to create a simple predictive model of archaeological site occurrence to assist in the design of future surveys in the Central Meseta, rather than develop a tool for interpreting patterns of prehistoric land-use. As a result, we use a small suite of environmental predictors, supplemented with climate variables.

We use logistic regression to model the probability of occurrence of archaeological remains using the presence/absence of archaeological finds in our survey region. We compared multiple candidate models using different combinations of climate and geographic variables (see below). The best model was selected from amongst the candidate models using Akaike’s information criterion (AIC) (Burnham & Anderson 2002) while also minimizing the variance inflation factor (VIF), a measure of correlation among predictor variables (Boyce et al. 2002).

Model performance was assessed using the receiver operator curve (ROC) and the associated metric, area under the curve (AUC). ROC analysis is a tool frequently used in predictive modelling to assess model performance and sensitivity (Boyce et al. 2002; Cullingham et al. 2012; Fielding and Bell 1997; Guisan and Zimmermann 2000; Jiménez-Valverde 2012). ROC curves are produced by plotting the false-positive rate against the true positive rate at different thresholds. The area under the ROC curve (the AUC) is a measure of overall model performance (i.e., the probability that the model is making correct predictions). A series of models were built

using 60 percent of the full data set and performance was assessed by comparing the predicted and observed values for the remaining 40 percent. Bootstrapped comparisons were repeated 9,999 times to generate a distribution of AUC values using different training data sets. Logistic regression was performed using the generalized linear models function in R, and ROC analysis was undertaken using the ROCR package (Sing et al. 2009).

The geographical predictors tested are elevation, measured using a 30 m digital elevation model (ASTER GDEMv2), slope, aspect (divided into four classes corresponding to cardinal points of the compass), and distance to water (using a cost-surface generated in ARCGIS 10 and the Hydro-1 Kilometer Watershed Model) with the presence or absence of lithic artefacts as the dependent variable. The extent to which existing hydrology (used to calculate the distance to water variable) is a good descriptor for Pleistocene conditions is unknown, but since our primary goal at this stage is to establish a predictive model based on existing conditions, rather than an interpretive model, we feel justified in including this parameter. Furthermore, the study region is characterised by a number of deeply incised rivers (e.g., the Tagus) whose current course is probably fairly representative of Late Pleistocene hydrology, though this remains to be demonstrated.

Two palaeoclimate predictors (average annual temperature and annual average daily precipitation) are also included in the list of predictors tested. The climate variables were obtained from a climate simulation produced by G. Levavasseur (*Laboratoire des Sciences du Climat et de l'Environnement*, Saclay-sur-Orme, France) for the Late Glacial Maximum (LGM). The Late Glacial

Maximum represents peak glacial conditions occurring between 19 and 22 kya (Yokoyama et al. 2000) that potentially affected the distribution of Palaeolithic populations in the central Meseta and is a good proxy for cold phases of marine isotope stage (MIS) 3 (Heinrich events), for which downscaled climate simulations are not yet available.

The rationale for including palaeoclimate predictors at this stage in the research is to test their potential use in conjunction with survey data for the future development of interpretive models of the study region. Additional data—i.e., securely dated archaeological sites—will be required before the full potential of these predictors can be realised. The palaeoclimate variables were produced by downscaling the results of a coupled atmosphere-ocean general circulation model (AOGCM), the IPSL_CM4. Previous studies (Vrac et al. 2007; Wilby and Wigley 2000; Wilby et al. 1998) show that statistical downscaling methods are a robust approach to generating climate variables at a local scale from climate models. The statistical downscaling was done following Vrac et al. (2007) using a Generalized Additive Model (GAM) that generates expectations for the explained variable (i.e., temperature or precipitation) from the climatology provided by the Climate Research Unit (CRU) as a function of geographical (e.g., topographical) or physical (i.e., simulated by IPSL-CM4) large-scale explanatory variables. GAM was performed in R (Wood 2006).

Results

Survey Results

The results of this initial phase of our research confirm the archaeological

potential of the target region. Almost without exception, the archaeological finds are concentrated in the western portion of the study region (Figure 2). The majority of the finds are situated either on bluffs above river valleys, primarily the Tagus and its tributaries, or on ancient river terraces. Seventy-three out of 198 locations surveyed (Table 1) yielded lithic artefacts for a total of 1,898 artefacts catalogued to date (Fuertes et al. 2009, 2010).

Localities with archaeological remains occur primarily on ancient river terraces, in sediments composed of cobbles, sandstones, clays and marls (e.g., CARRA,

see below), and on bluffs above the river in detrital sediments resulting from the weathering of local bedrock (primarily composed of sandstone). None of the caves surveyed to date, and only one rockshelter, yielded artefacts.

Lithic Analysis

Typological and technological analyses of the three largest lithic assemblages (Figure 3) were undertaken in order to determine the range of variation of archaeological materials encountered and establish an initial chronological framework for the archaeological record of the target region. Preliminary analyses

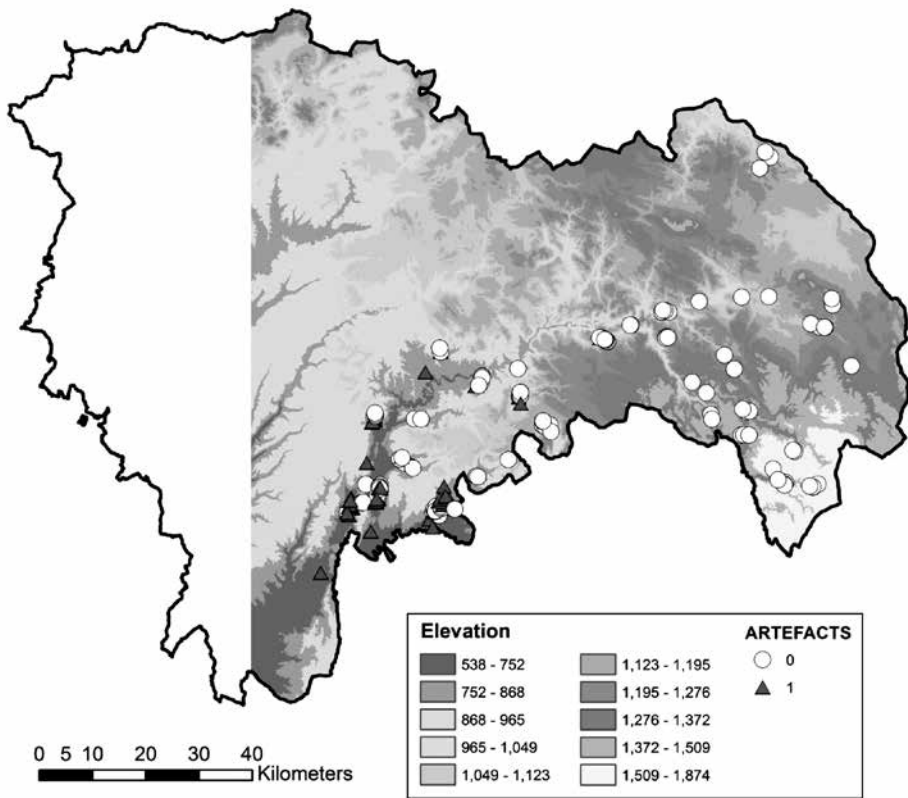


FIGURE 2. Survey locations: localities with and without lithic artefacts are mapped against the cropped digital elevation model (DEM) indicating the portion of the study region used in the spatial analyses.

TABLE 1. Results of the 2009 and 2010 surveys, by municipality (see Figure 1); the number of survey locations (centroids mapped in Figures 2 and 3) and localities with lithic artefacts (“findspots”) are indicated.

Municipality	Findspots	Number of Survey	
		Locations	% Findspots
Alcocer	10	22	45.5
Arbeteta	2	13	15.4
Auñón	15	21	71.4
Budia	7	11	63.6
Buendia	1	1	100
Cifuentes	0	1	0
Cubillejo de la Sierra	0	1	0
El Recuenco	0	4	0
Fuentelsaz	0	3	0
Labros	0	1	0
Lebrancón	0	14	0
Mantiel	0	2	0
Molina de Aragon	0	1	0
Orea	0	12	0
Pareja	6	11	54.5
Peralejos de las Truchas	0	9	0
Peralveche	1	3	33
Prados Redondos	0	7	0
Sacedón	28	34	83
Setiles	0	1	0
Taravilla	0	5	0
Trillo	2	3	66.7
Valdeolivas	0	3	0
Valhermoso	0	1	0
Zaoreja	1	14	7
Total	73	198	

of the three assemblages, designated as archaeological sites (Fuentes et al. 2009, 2010), were undertaken by M. Bisson (see below). These assemblages were selected for analysis because of their size and apparent typological and technological homogeneity.

The raw material used at all three locations is almost exclusively local, occurring as flint nodules commonly encountered near the sites. Pseudo-retouch is common and could result from either geological processes or agricultural activities leading to sheet erosion (Barton et al. 1999). Most of the pieces are dehydrated, which would have made them relatively fragile and

more apt to suffer post-depositional damage.

Typologically, the lithic artefacts indicate occupations spanning from the early Middle Palaeolithic (CARRA, see below) to the Middle and Upper Palaeolithic, including recent Prehistory (i.e., Neolithic or Chalcolithic).

CARRA (“*La Carrascosilla*”). CARRA is located approximately 3 km west of Sacedón, on alluvial terraces on the left bank of the Tagus River (more precisely, the Embalse de Bollarque). The assemblage (Figure 4) is primarily composed of flakes and flake fragments (Figure 4b). Thirty formal tools (9.2

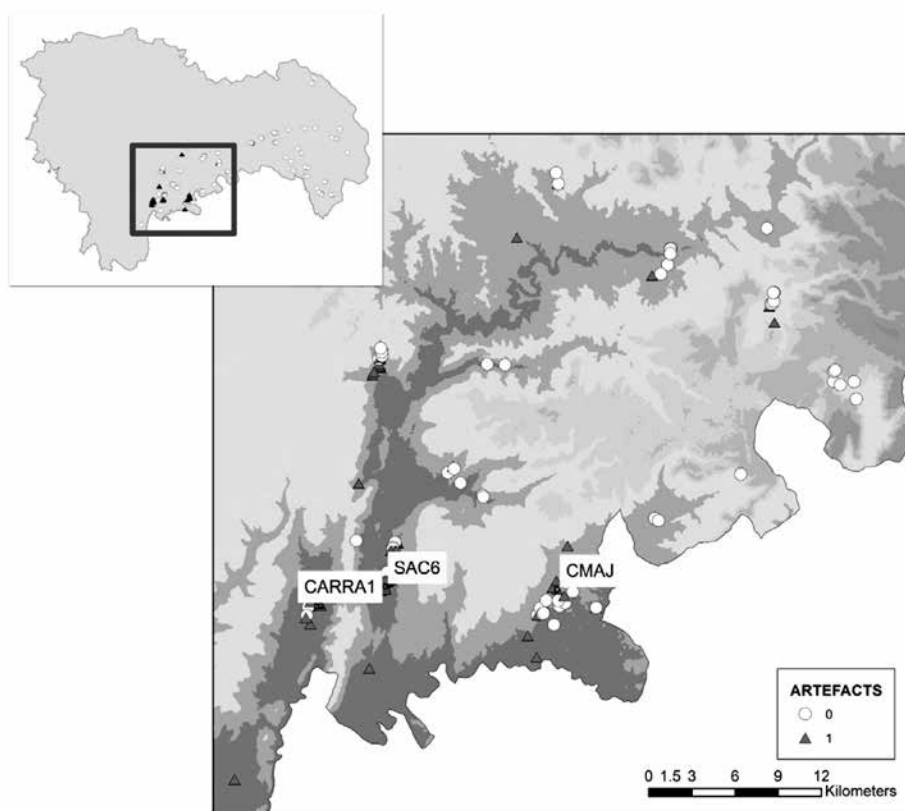


FIGURE 3. The three localities chosen for detailed analysis: CARRA “las Carrascosilla,” SAC6 “Sacedón 6,” and CMAJ “Camino de las Majadillas.” All three lithic assemblages were considered relatively large and typologically homogeneous.

percent of the total) were identified including notches (23 percent), denticulates (16.7 percent), Levallois flakes, endscrapers on flake blanks, sidescrapers (13.3 percent), a borer, a Mousterian point fragment, and a possible Mousterian-cleaver (Figure 4c). The 53 cores in this assemblage are predominantly informal types, many of which (20.8 percent) are on very large, thick flakes, the interior surface of which served as the core striking platform (Figure 4a). Given the techno-typological composition of the assemblage, we provisionally suggest that it could represent an Early Middle Palaeolithic occupation.

SAC6 (*Sacedón 6*). SAC 6 is located about 1.5 km north of the village of Sacedón, in a swale between two parallel sandstone ridges at the top of a hill overlooking the Entrepeñas reservoir. Flakes and flake fragments compose 60 percent of this collection of artefacts ($N=956$), but blades, many of which were obtained by hard percussion (Figure 5a), also occur in appreciable numbers (ca. 14 percent). Cores (6.4 percent) are relatively rare despite the fact that the site is close to an outcrop of good chert. Formal tools occur in moderate numbers (4.6 percent) (Figure 5c). Single and double platform cores intended for the produc-

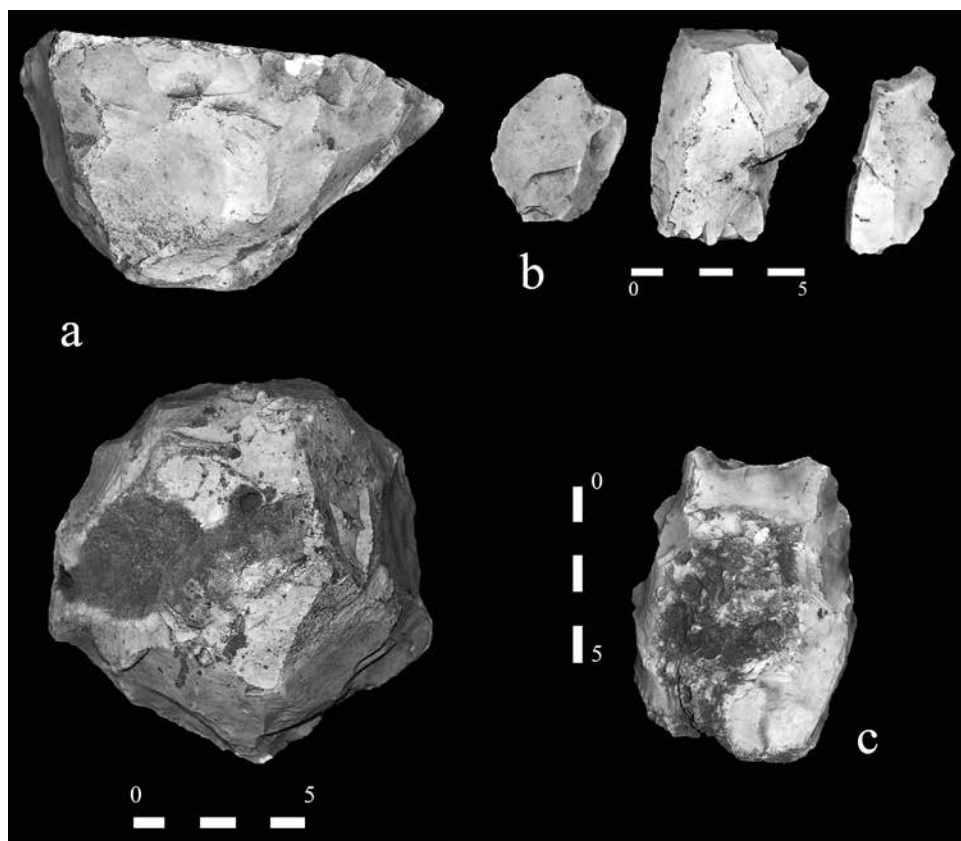


FIGURE 4. CARRA (“las Carrascosilla”) lithics: (a) large atypical *rabot*/single platform core; (b) unmodified flakes; (c) Vasconian cleaver with a concave bit.

tion of elongated flakes dominate, but blade and bladelet cores constitute ca. 20 percent of the collection (Figure 5d–e). There is only one Levallois flake core. The SAC6 collection is clearly a palimpsest, incorporating a Middle Palaeolithic component (presence of flakes, *éclats débordants*, and laminar flakes produced using hard hammer percussion) and a numerically more important Upper Palaeolithic component with blades and bladelets produced by soft percussion (Figure 5b). Upper Paleolithic formal tools include simple endscrapers, endscrapers on retouched blades, a truncated blade and a backed bladelet.

A post-Palaeolithic component may also be present (possibly Chalcolithic or Bronze Age). The topography of the site indicates that it is a sediment trap; it is therefore possible that there are multiple buried components in SAC6 which warrant further investigation.

CMAJ (“Camino Majadillas”). CMAJ is located just over 12 km east-northeast of Sacedón on a low ridge between the Arroyo de la Fuente Gris and the Arroyo del Tejar, which drain into the Río Garrigay. The ridge, along which a dirt road runs, is covered by woods and scrub vegetation on one side and is cultivated on

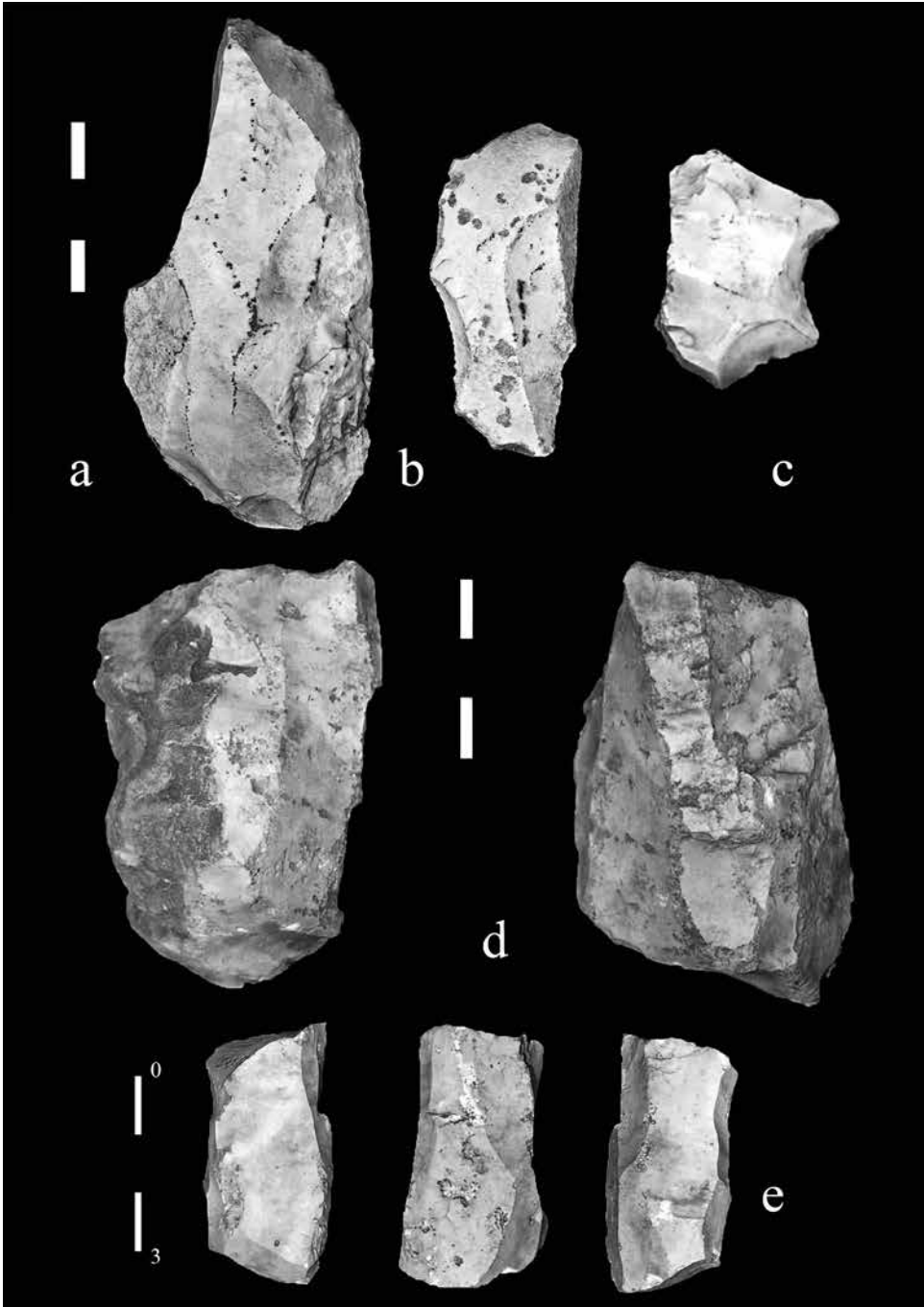


FIGURE 5. SAC6 (“Sacedón 6”) lithics: (a) elongated flake with a distal burination; (b) blade; (c) compound notch on a flake; (d) blade core; (e) prismatic bladelet core.

the other. A 25 m wide transect on either side of the road was surveyed for the full length (ca. 500 m) of the ridge yielding 249 artefacts, mostly from the wooded side of the ridge.

All of the diagnostic tools and most of the cores are Middle Palaeolithic types and technological attributes also point to the Middle Palaeolithic. Flakes and flake fragments dominate (75.5 percent); the total of formal tools is 12.4 percent (Figure 6b, d, e) and notches (35 percent) and denticulates (30 percent) dominate the retouched pieces (Figure 6c). Levallois cores are the most common, followed by casual forms (Figure 6a). Given the relatively high frequency of formal tools in this assemblage, it is possible that this locality represents a palimpsest of Middle Palaeolithic occupations on a ridge that offered a panoramic view of the surrounding countryside.

In summary, there is a clear Middle Palaeolithic signature in the lithic assemblages examined to date, based on the production of flakes using hard hammer percussion and the use of Discoid and Levallois technology, as well as the dominance of retouched pieces, notches and denticulates among the tools. It is possible that the CARRA assemblage corresponds to the Early Middle Palaeolithic, as indicated by the presence of a cleaver and cores on large flakes that can be classified typologically as atypical *rabots*. CMAJ is characteristic of the Middle Palaeolithic on typological and technological grounds and a Middle Palaeolithic component is present at SAC 6. However, the bulk of the assemblage from SAC 6 is attributable to the Upper Palaeolithic, given the relatively high percentage of blades and bladelets produced by soft percussion.

Spatial Analysis

Having established that the target region has considerable archaeological potential, the next step in our research programme was the construction of a spatial model to describe the pattern of archaeological occurrences quantitatively using logistic regression.

The best model for the prediction of archaeological occurrences (Table 2: model 6) includes elevation and slope as predictors. Model performance of Model 6 was assessed using bootstrapped estimates of the area under the ROC curve (AUC). Using 9,999 bootstrapped comparisons involving 60 percent and 40 percent testing data, we calculated a mean AUC of 0.8445 (SE = 0.00034) which indicates that the selected model provides a correct prediction of archaeological potential 84 percent of the time. This model can now be applied to the full extent of the target region (eastern Guadalajara province) to produce an initial predictive model of archaeological potential (Figure 7).

Two palaeoclimate variables (average annual temperature and precipitation during the LGM) were included as predictors in a separate run of models. Although average precipitation was quickly discarded as a predictor, average annual temperature proved to be a good predictor (Figure 8). Temperature is correlated with elevation, however, which is already included in our logistic model. This correlation is to be expected given empirical data (temperature gradients increase with increased elevation) and the methodology used to produce the downscaled climate results (the GAM incorporates topography). Including climate predictors at this stage was a useful exercise, however, since it demonstrated their potential. We hope that future fieldwork will yield sufficient

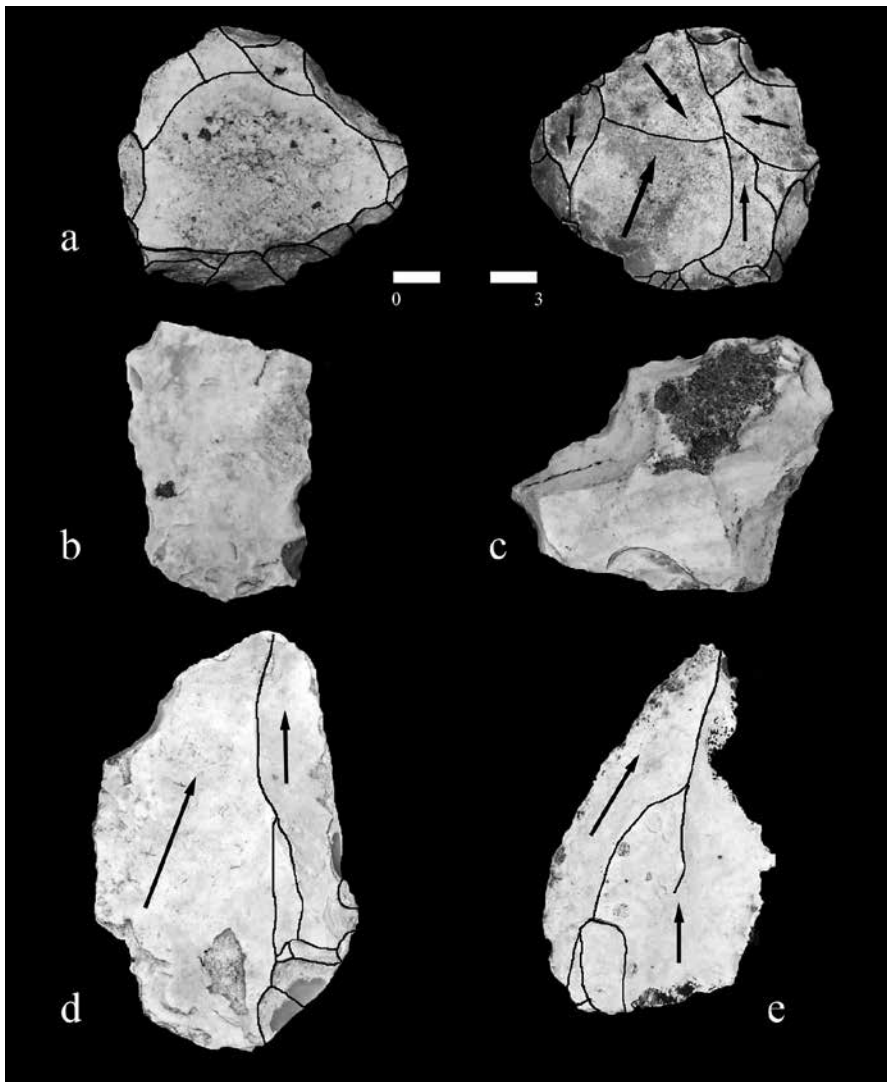


FIGURE 6. CMAJ (“Camino Majadillas”) lithics: (a) centripetal recurrent Levallois core; (b) proximal end of a broken Levallois blade; (c) denticulate; (d) elongated Levallois flake; (e) pseudoLevallois point (notch on distal right margin is probably post-depositional damage).

archaeological data to enable us to test the full range of predictors and develop interpretive models of patterns of land use in the Central Meseta during the Late Pleistocene. We are currently working on a downscaled climate simulation for MIS 3 as well as an expanded survey.

Discussion

Visual assessment of the distribution of survey locations suggests that archaeological finds occur at lower elevations (Figure 2). The predictive model (above), which includes elevation and slope as predictors, confirms this impres-

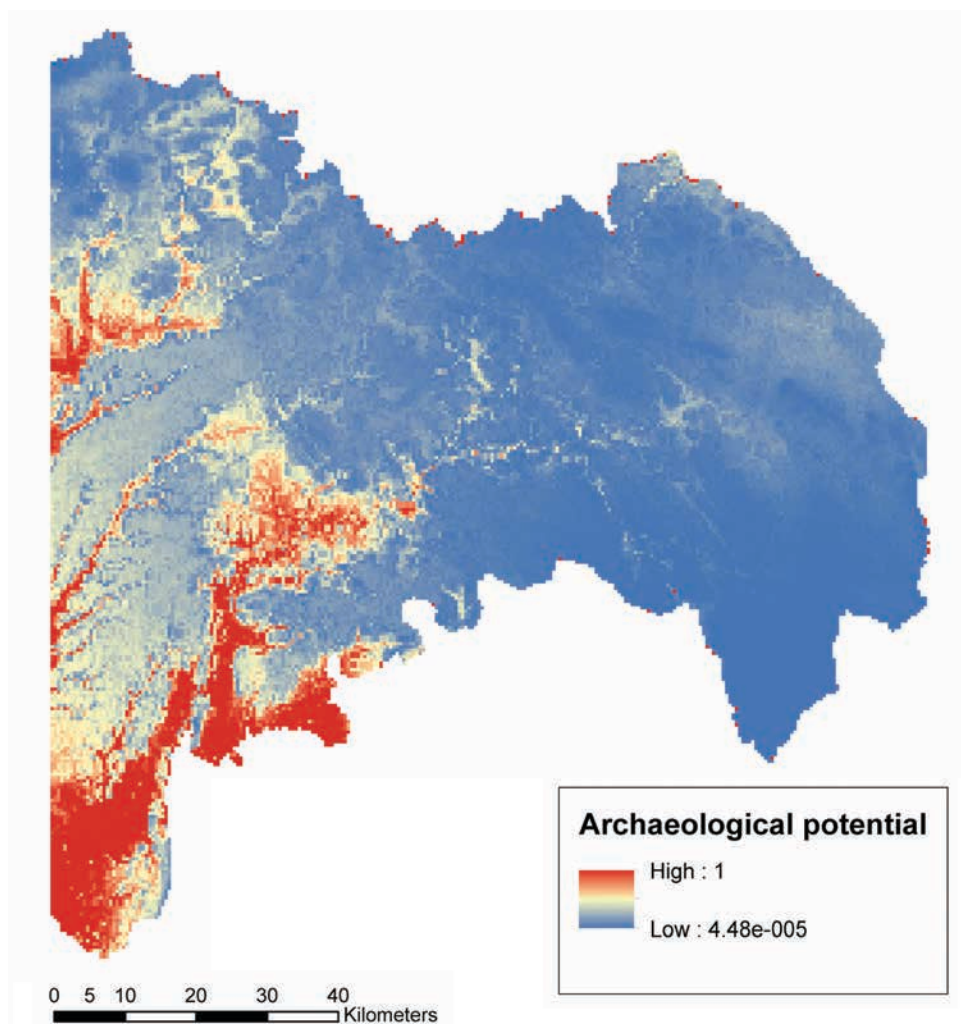


FIGURE 7. The archaeological predictive model for eastern Guadalajara province. “Hot” colours indicate high potential for the discovery of surface finds.

sion (Table 2). The mean elevation of survey locations with artefacts is 751 m above sea level (asl) (min: 619 m asl; max: 1,225 m asl) compared with a mean elevation of 1,092 m asl (min: 538 m asl; max: 1,874 m asl) for the survey region. The mean slope value for locations with artefacts is 6.34 degrees (min: 1.34 degrees; max: 25.22 degrees) compared with a mean of 7.3 degrees for the study region (min: 0 degrees; max:

63.3 degrees). There is also an apparent trend in the distribution of archaeological remains in relation to aspect, although this variable is not a significant predictor (Table 2). Sixty-four percent of the survey localities with artefacts are encountered in the southern quadrant (i.e., south, south-east, or south-west aspects) and only 12 percent are located in the northeast quadrant, compared with equal distributions of cells by aspect

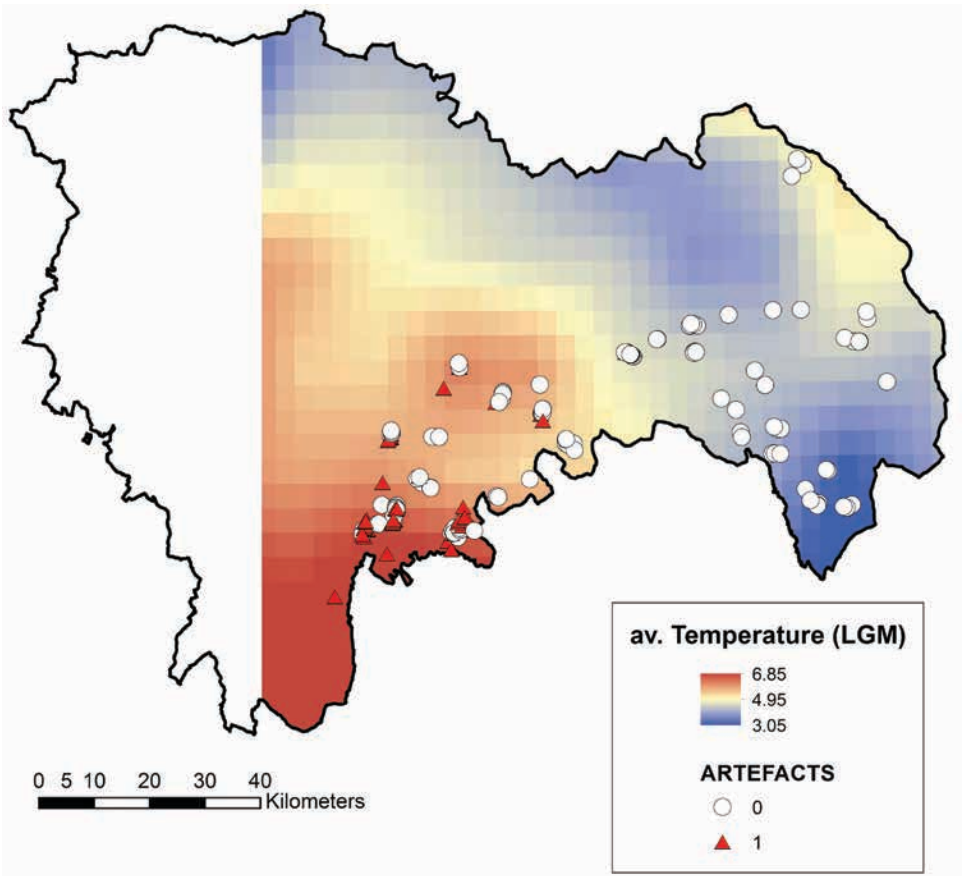


FIGURE 8. Survey locations mapped against a backdrop of average annual temperatures for the Late Glacial Maximum (LGM). Similar temperature gradients are observable today and are correlated with altitudinal gradients.

TABLE 2. Model selection using Aikake’s information criterion (AIC) and cross-validation results for Model 6 using the Receiver Operating Characteristic (ROC) and Area Under Curve (AUC).

Model	Variables	AIC	δ AIC
1	Elevation	177.58	7.2
2	Slope	223.26	52.9
3	Distance to Water	241.57	71.2
4	Aspect	263.71	93.3
5	Elevation + Distance to Water	179.58	9.2
6	Elevation + Slope	170.37	0
7	Slope + Distance to Water + Elevation	171.23	0.85
8	Aspect Class + Slope + Elevation	172.09	1.72

Note: Model 6 AUC: 0.8445 (SE=0.00034)

in the regional environment. These data suggest a bias towards sunnier locations, although this needs to be tested further.

The predictive model generated using Model 6 (Figure 7) will be used to design future surveys of the study region (e.g., in the municipalities not yet visited to date). The model also highlights differences between the western and eastern portions of the study region, which yielded significantly different survey results, and raises an interesting question: i.e., whether different taphonomic processes are shaping the archaeological record of the western and eastern portions of the study region. For example, sheet erosion may have occurred more frequently in the western portion of the study region resulting in surface concentrations of archaeological remains in deflated terraces and sediment traps (e.g., Barton et al. 1999; Barton et al. 2002). If this proves to be the case, our survey strategy will have to be modified accordingly. Plans for improving upon the spatial model in the future in order to address this question include adding geological variables to the list of predictors (geological maps at 1:25,000 scale do not use consistent notation and we are currently working to resolve this). We also plan to incorporate soil erosion modelling (RUSLE3D and USPED) into the spatial analysis to test whether erosion is contributing to the observed spatial patterning.

Our palaeoclimate predictors hinted at another possibility: temperature (which correlates with elevation) is not equally distributed across the study region demonstrating a distinct east/west gradient. This suggests that climate conditions, particularly temperature during cold phases of the Late Pleistocene, could have had an impact on the pattern of human occupation, as sug-

gested by several authors for the Central Meseta more generally (see above). We hope to be in a position to investigate this hypothesis in the near future, using a combination of survey results and palaeoclimate variables.

Conclusion

The surveys undertaken during the summers of 2009 and 2010 have yielded evidence of previously undocumented Palaeolithic occupation in the eastern portion of Guadalajara province. Chrono-cultural attribution of the lithic remains encountered is difficult due to the fact that open-air deposits in the target region probably represent palimpsests. Preliminary analysis of the three largest lithic assemblages recovered, however, indicates that the target region preserves a long history of occupation, from the early Middle Palaeolithic to proto-historic periods (Fuentes et al. 2009, 2010).

Archaeological remains were only encountered in the western part of the target region, in open-air contexts on river terraces or on bluffs above river valleys. The predictive model derived from these results suggests that elevation and slope are the best predictors of archaeological potential for the region. There are two possible explanations for the spatial patterning of the archaeological remains, however. As discussed above, there is a possibility that the western and eastern portions of the study region are affected by different taphonomic processes which could account for differential preservation of archaeological remains. This hypothesis will be evaluated in the future by testing additional variables such as sediment type, as well as modelling erosion processes. However, initial tests of palaeoclimate variables (annual average

temperature and precipitation) suggest an alternative hypothesis, namely that the spatial patterning of the archaeological remains reflects differences in climate, linked to topography and potentially to human selection. We are currently expanding our palaeoclimate database in order to explore this issue further.

A planned programme of archaeological test-pitting in the larger caves identified during our survey, all of which are in the eastern portion of the study region, will also enable us to test the existing predictive model. With further field surveys and more sophisticated spatial modelling, including the addition of new palaeoclimate variables, we hope to be in a position to address some of the important issues that have been raised concerning the pattern of Palaeolithic occupation in the Central Meseta and its impact on the history of human settlement in the Iberian Peninsula.

Acknowledgements. The authors would like to thank the following people without whose hard work and dedication the fieldwork upon which this article is based would not have been possible: Dr. Carlos Arteaga-Cardineau (Profesor Ayudante, Departamento de Geografía de la Universidad Autónoma de Madrid), D. Guiducci and L. Bourgeon (Doctoral programme, Université de Montréal). We also thank three anonymous reviewers whose constructive criticism was very helpful in revising the original manuscript.

This article forms part of a special issue in honour of Michael Bisson, whose energy and enthusiasm made a significant contribution to the fieldwork that forms the basis of this article. His co-authors would all like to take this opportunity to thank him for his hard work, his devotion to the project, and his friendship.

References Cited

- Aguirre, Emiliano, and Eudald Carbonell
2001 Early Human Expansions into Eurasia: The Atapuerca Evidence. *Quaternary International* 75:11–18.
- Alcolea González, J.J., and R. de Balbín Behrmann
2003 Témoins du froid. La faune dans l'art rupestre paléolithique de l'intérieur péninsulaire. *L'Anthropologie* 107:471–500.
- Alcolea González, José Javier, Rodrigo de Balbín, Miguel Angel García, Pedro José Jiménez, Amparo Aldecoa, Ana Belén Casado, Benito de Andrés, Susana Ruíz, Pilar Sáinz, and Natalia Suárez
1998 Avance al estudio del poblamiento del alto valle del Sorbe (Muriel, Guadalajara). *Proceedings of the 2 Congreso de Arqueología peninsular* 1:201–218.
- Arribas, A., J.C. Díez, and Jesús F. Jordá
1995 El yacimiento cuaternario de la Cueva de Los Torrejones (Tamajón, Guadalajara, Castilla-La Mancha): resultados preliminares. In *Arqueología en Guadalajara*, edited by Rodrigo de Balbín, Jesús Valiente Malla, and Maria Teresa Mussat, pp. 97–110. Patrimonio Histórico-Arqueología Vol. 12. Servicio de Publicaciones de la Junta de Comunidades de Castilla-La Mancha, Castilla-La Mancha, Spain.
- Arsuaga, J.L., E. Baquedano, and A. Pérez-González
2011 Neanderthal and Carnivore Occupations in Pinilla del Valle (Community of Madrid, Spain). In *Proceedings of the XV UISPP World Congress (Lisbon, 4–9 September 2006)*, edited by Luiz Oosterbeek and Cláudia Fidalgo, pp. 111–119. BAR International Series 2224. Archaeopress, Oxford.

Baena, Javier, José Polo, Sergio Báñez, Felipe Cuartero, Marta Roca, Ana Lázaro, Ana Nebot, Alfredo Pérez-González, Teresa Pérez, Inmaculada Rus, Daniel Rubio Ruiz, Diego Martín-Puig, Carmen Manzano, Iván González, and Raúl Márquez

2008 Tecnología musteriense en la región madrileña: un discurso enfrentado entre valles y páramos de la Meseta Sur. *Treballs de Arqueologia* 14:249–278.

Balbín Berhmann, R., J.J. Alcolea González, F. Moreno Sanz, and L.A. Cruz Naimi

1995 Investigaciones arqueológicas en la Cueva de La Hoz (Santa María del Espino, Guadalajara). Una visión de conjunto actualizada. In *Arqueología en Guadalajara*, edited by Rodrigo de Balbín, Jesús Valiente Malla, and Maria Teresa Mussat, pp. 37–53. Patrimonio Histórico-Arqueología Vol. 12. Servicio de Publicaciones de la Junta de Comunidades de Castilla-La Mancha, Castilla-La Mancha, Spain.

Barandiarán, Ignacio Maestro

1973 *La Cueva de los Casares en Riba de Saelves, Guadalajara*. Excavaciones arqueológicas en España Vol. 76. Ministerio de Educación y Ciencia, Dirección General de Bellas Artes, Madrid, Spain.

Barton, C. Michael, Joan Bernabeu, J. Emili Aura, and Oreto García

1999 Land-Use Dynamics and Socio-economic Change: An Example from the Polop Alto Valley. *American Antiquity* 64:609–634.

Barton, C. Michael, Joan Bernabeu, J. Emili Aura, Oreto García, and Neus La Roca

2002 Dynamic Landscapes, Artifact Taphonomy, and Landuse Model-

ing in the Western Mediterranean.

Geoarchaeology: An International Journal 17:155–190.

Beltrán, Antonio, and Ignacio Barandiarán

1968 *Avance al estudio de las cuevas paleolíticas de La Hoz y Los Casares (Guadalajara)*. Excavaciones Arqueológicas en España Vol. 64. Ministerio de Educación y Ciencia, Dirección General de Bellas Artes, Madrid, Spain.

Bevan, Andrew, and James Conolly

2002 GIS, Archaeological Survey, and Landscape Archaeology on the Island of Kythera, Greece. *Journal of Field Archaeology* 29:123–138.

Boyce, Mark S., Pierre R. Vernier, Scott E. Nielsen, and Fiona K.A. Schmiegelow

2002 Evaluating Resource Selection Functions. *Ecological Modelling* 157:281–300.

Cacho, C., J.A. Martos, J.J. Jordá, J. Yravedra, B. Avezuela, J. Valdivia, and I. Martín

2010 El Paleolítico superior en el interior de la Península Ibérica. Revisión crítica y perspectivas de futuro. In *El Paleolítico Superior Peninsular: Novedades del Siglo XXI*, edited by Xavier Mangado, pp. 75–96. Monografies del Seminari d'Estudis i Recerques Prehistòriques, Barcelona, Spain.

Carbonell, E., J.M. Bermúdez de

Castro, J.L. Arsuaga, J.C. Díez, A. Rosas, G. Cuenca-Bescós, R. Sala, M. Mosquera, and X.P. Rodríguez

1995 Lower Pleistocene Hominids and Artifacts from Atapuerca-TD6 (Spain). *Science* 269:826–830.

Chapa, Teresa

2006 Arte Paleolítico. In *Prehistoria y Protohistoria de la Meseta Sur (Castilla-La*

- Mancha*), edited by Juan Pereira Sieso, pp. 41–48. Biblioteca Añil Vol. 31. Almad, Ciudad Real, Castile-La Mancha, Spain.
- Corchón, Soledad, José Antonio Abásolo, Julián Bécares, Javier Gonzalez-Tablas, Marcos Mayer, Tomás Romero, and Carmen Sevillano
1997 La Cueva de la Griega de Pedraza (Segovia). Junta de Castilla y León, Arqueología en Castilla y León, *Memoorias* 3:29–181.
- Cullingham, Catherine I., Patrick M. A. James, Janice E. K. Cooke, and David W. Coltman
2012 Characterizing the Physical and Genetic Structure of the Lodgepole Pine × Jack Pine Hybrid Zone: Mosaic Structure and Differential Introgression. *Evolutionary Applications* 5:879–891.
- Dalla Bona, L.
1994 *Cultural Heritage Resource Predictive Modeling Project: Vol. 3 Methodological Considerations*. Centre for Archaeological Resource Prediction, Thunder Bay, Ontario.
- de la Torre, Ignacio
2007 La ocupación paleolítica de la Meseta Sur. In *Prehistoria y Protohistoria de la Meseta Sur (Castilla-La Mancha)*, edited by Juan Pereira Sieso, pp. 17–39. Almad, Ciudad Real, Castile-La Mancha, Spain.
- de la Torre, Ignacio, Jorge Martinez-Moreno, Elías Lopez-Romero, John Gowlett, Noemí Morán, Juan M. Vincent, and Alfonso Benito
2007 Primeras intervenciones arqueológicas en el yacimiento paleolítico del abrigo de Buendía (Castejón, Cuenca). In *Arqueología de Castilla-la Mancha*, edited by Juan Manuel Millán Marínez and Concepción Rodríguez Ruza, pp. 531–543. Servicio de Publicaciones de la Universidad de Castilla-la Mancha, Cuenca, Ecuador.
- Díez-Martin, Fernando, Policarpo Sánchez-Yustos, José Angel Gómez-González, and Diana Gómez de la Rúa
2008 Earlier Palaeolithic Settlement Patterns: Landscape Archaeology on the River Duero Basin Plateau (Castilla y León, Spain). *Journal of World Prehistory* 21:103–137.
- Ebert, David
2004 Applications of Archaeological GIS. *Canadian Journal of Archaeology/ Journal Canadien d'Archéologie* 28:319–341.
- Fernández Tabera, Miguel, and Pedro Martín Yebra
1982 *Catálogo de cavidades de Guadalajara*. Federación Castellana Centro de Espeleología, Madrid, Spain.
- Fielding, Alan H., and John F. Bell
1997 A Review of Methods for the Assessment of Prediction Errors in Conservation Presence/Absence Models. *Environmental Conservation* 24:38–49.
- Fuertes, Natividad, Arianne Burke, and José Maíllo Fernández
2009 *Cambio climático y dinámica de población durante la transición Paleolítico Medio-Superior de la Península Ibérica: el ejemplo del sur de la provincia de Guadalajara*. Informe sobre el permiso de intervención arqueológica (prospección) no. 091201. Available at the Consejería de Cultura, Turismo y Artesanía de la Junta de Comunidades de Castilla-La Mancha.
- 2010 *Cambio climático y dinámica de población durante la Transición Paleolítico Medio-Superior de la Península Ibérica: el ejemplo del sur de la provincia de Gua-*

- dalajara, fase II. campaña 2010. Informe sobre el permiso de intervención arqueológica (prospección) no. 101203.* Available at the Consejería de Cultura, Turismo y Artesanía de la Junta de Comunidades de Castilla-La Mancha.
- Gaffney, V., and P. M. van Leusen
1995 Postscript—GIS, Environmental Determinism and Archaeology: A Parallel Text. In *Archaeology and Geographical Information Systems: A European Perspective*, edited by G. Gary R. Lock and Zoran Stančić, pp. 367–382. Taylor and Francis, London.
- García Valero, Miguel Angel
1997 Aproximación al Paleolítico Medio en la vertiente sur del Sistema Central: Guadalajara. *Proceedings of the Actas del II Congreso de Arqueología Peninsular* 1:85–104. Zamora, Spain.
- García, Rogelio Estrada, Jesús F. Jordá Pardo, and Javier Pastor Muñoz
1992 El yacimiento prehistórico de Jarama I (Valdesotos, Guadalajara). Campaña de prospección de 1991. *Espacio, Tiempo y Forma, Serie I, Prehistoria y Arqueología* 5:153–164.
- Guisan, Antoine, and Niklaus E. Zimmermann
2000 Predictive Habitat Distribution Models in Ecology. *Ecological Modelling* 135:147–186.
- Jiménez-Valverde, Alberto
2012 Insights into the Area Under the Receiver Operating Characteristic Curve (AUC) as a Discrimination Measure in Species Distribution Modelling. *Global Ecology and Biogeography* 21:498–507.
- Jimeno Martínez, Alfredo, José Javier Fernández Moreno, Juan Antonio Gómez-Barrera, and María Pilar Galindo Ortiz de Landázuri
1990 Arte paleolítico en la provincia de Soria: La placa de Villalba. *Numantia* 3:9–50.
- Jordá Pardo, Jesús F., and Miguel Angel García Valero
1989 Las representaciones de glotón (*Gulo gulo* L.) en el arte paleolítico pirenaico y un nuevo hallazgo de arte mueble en el alto valle del Jarama (Guadalajara). *Espacio, Tiempo y Forma, Serie I, Prehistoria y Arqueología* 2:89–107.
- Kvamme, Kenneth L.
2006 There and Back Again: Revisiting Archaeological Locational Modeling. In *GIS and Archaeological Site Location Modeling*, edited by Mark W. Mehrer and Konnie L. Wescott, pp. 2–40. Taylor & Francis, Boca Raton, Florida.
- López-García, Juan Manuel, Hugues-Alexandre Blain, Gloria Cuenca-Bescós, María Blanca Ruiz-Zapata, Miriam Dorado-Valiño, María José Gil-García, Ana Valdeolmillos, Ana Isabel Ortega, Jose Miguel Carretero, Juan Luis Arsuaga, Jose María Bermúdez de Castro, and Eudald Carbonell
2010 Palaeoenvironmental and Palaeoclimatic Reconstruction of the Latest Pleistocene of El Portalón Site, Sierra de Atapuerca, Northwestern Spain. *Palaeogeography, Palaeoclimatology, Palaeoecology* 292(3–4):453–464.
- McCoy, Mark D., and Thegn N. Ladefoged
2009 New Developments in the Use of Spatial Technology in Archaeology. *Journal of Archaeological Research* 17:263–295.
- Malla, Jesús Valiente
1997 *Guía de la arqueología en Guadalajara*. Tierra de Guadalajara Series Vol. 15. AACHE Ediciones de Guadalajara, Guadalajara, Jalisco, Mexico.

Mercado Blanco, Jesús, María Jesus Moya Benito, and Antonio Herrera Casado

2003 *Historia de Sacedón: Patrimonio y costumbres*. AACHE Ediciones de Guadalajara, Guadalajara, Jalisco, Mexico.

Mosquera, Marina, Andreu Ollé, Alfredo Pérez González, Xose Pedro Rodríguez, Manuel Vaquero, Josep María Vergès, and Eudald Carbonell

2007 Valle de las Orquídeas: un yacimiento al aire libre del Pleistoceno Superior en la Sierra de Atapuerca (Burgos). *Trabajos de Prehistoria* 64(2):143–155.

Mosquera, M., A. Ollé, and X.P. Rodríguez

2013 From Atapuerca to Europe: tracing back the earliest peopling of Europe. *Quaternary International*, in press.

Moure Romanillo, José Alfonso, and Ernesto García-Soto Mateos

1983 Cueva Millán y la Ermita. Dos yacimientos musterianos en el valle medio del Arlanza. *Boletín del Seminario de Estudios de Arte y Arqueología* 49:5–30.

Navazo, Marta, and J. Carlos Díez

2008 Prado vargas y la variabilidad tecnológica a finales del paleolítico Medio en la meseta norte. *Treballs d'Arqueologia* 14:121–139.

Navazo, M., J. C. Díez, T. Torres, A. Colina, and J. E. Ortiz

2005 La cueva de Prado Vargas. Un yacimiento del Paleolítico Medio en el sur de la Cordillera Cantábrica. In *Neandertales cantábricos, estado de la cuestión*, edited by Ramón Montes Barquín and José Lasheras Corrucho, pp. 151–166. Monografías Vol. 20. Museo Nacional y Centro de Investigación de Altamira, Santillana del Mar, Cantabria, Spain.

Panera Gallego, J.

2010 La ocupación del medio fluvial en el Paleolítico antiguo: Caracterización geoarqueológica de depósitos pleistocenos del Valle del Río Jarama (Madrid) y estudio tecnoeconómico de la industria lítica. Unpublished Ph.D. dissertation, Departamento de Prehistoria y Arqueología, Universidad Nacional de Educación a Distancia, Madrid, Spain.

Rasilla Vives, Marc, Code la, Manuel Hoyos Gómez, and Juan Carlos Cañaveras Jiménez

1996 El abrigo de Verdelpino (Cuenca). Revisión de su evolución sedimentaria y arqueológica. *Complutum Extra* 6:75–82.

Ripoll, S., and Y. L. J. Municio

1999 Domingo García Arte Rupestre Paleolítico al Aire Libre en la Meseta Castellana. Junta de Castilla y León, Arqueología en Castilla y León, Memorias 8:278.

Rodríguez, Jesús

2004 Stability in Pleistocene Mediterranean Mammalian Communities. *Palaeogeography, Palaeoclimatology, Palaeoecology* 207:1–22.

Rubio Jara, Susana

2011 El Paleolítico en el valle del río Manzanares (Madrid): caracterización geoarqueológica de depósitos pleistocenos y estudio tecnoeconómico de la industria lítica. Unpublished Ph.D. dissertation, Departamento de Prehistoria y Arqueología, Universidad Nacional de Educación a Distancia, Madrid, Spain.

Sánchez, Policarpo, Fernando Díez Martín, Isabel M. Díaz Muñoz, Diana Gómez de la Rúa, and José Angel Gómez González

2011 Estrategias de talla en Cueva

- Corazón (Mave, Palencia). Un yacimiento del Musteriense Antiguo en las estribaciones meridionales de la Cordillera Cantábrica. *Trabajos de Prehistoria* 68(1):51–63.
- Santonja, Manuel, and Alfredo Pérez González
2000/2001 El Paleolítico Inferior en el interior de la Península Ibérica: un punto de vista desde la geoarqueología. *Zephyrus* 53–54:27–77.
2005 *Los yacimientos paleolíticos de Ambrona y Torralba (Soria): un siglo de investigaciones arqueológicas*. Zona Arqueológica No. 5. Museo Arqueológico Regional, Alcalá de Henares, Madrid, Spain.
- Sing, Tobias, Oliver Sander, Niko Beerenwinkel, and Thomas Lengauer
2009 Visualizing the Performance of Scoring Classifiers. In *Package ROCR Version 1.0-4*.
- Utrilla, P., L. Montes, C. Mazo, A. Alday, J. M. Rodanés, M. F. Blasco, R. Domingo, and M. Bea
2010 El Paleolítico superior en la cuenca del Ebro a principios del siglo XXI. Revisión y novedades. *El Paleolítico Superior Peninsular. Novedades del Siglo XXI*:23–62. Barcelona, Spain.
- Vega Toscano, Luis Gerardo, and Pedro Martín Blanco
2006 Análisis preliminar de las cadenas operativas en el material lítico procedente del nivel IV del Abrigo del Palomar (Yeste, Albacete). *Zona Arqueologica* 7(1):397–404.
- Vega Toscano, Luis Gerardo, Paloma Sevilla García, Fernando Colino Polo, Paloma de la Peña Alonso, Rosa Rodríguez Herranz, Fernando Gutiérrez Martín, and Sergio Báez
2010 Nuevas investigaciones sobre los yacimientos paleolíticos de la Sierra Norte de la Comunidad de Madrid. *Actas de las Quintas Jornadas de Patrimonio Arqueológico de la Comunidad de Madrid*:115–132. Madrid, Spain.
- Vrac, M., D. Paillard, and P. Naveau
2007 Non-Linear Statistical Downscaling of Present and LGM Precipitation and Temperatures Over Europe. *Climate of the Past* 3:899–933
- Wheatley, David, and Mark Gillings (editors)
2004 *Spatial Technology and Archaeology: The Archaeological Applications of GIS*. Taylor & Francis, New York.
- Wilby, R. L., and T. M. L. Wigley
2000 Precipitation Predictors for downscaling: Observed and General Circulation Model Relationships. *International Journal of Climatology* 20:641–661.
- Wilby, R. L., T. M. L. Wigley, D. Conway, P. D. Jones, B. C. Hewitson, J. Main, and D. S. Wilks
1998 Statistical Downscaling of General Circulation Model Output: A Comparison of Methods. *Water Resources Research* 34:2995–3008.
- Wood, Simon
2006 *Generalized Additive Models: An introduction with R*. Chapman and Hall/CRC, Boca Raton, Florida.
- Yokoyama, Yusuke, Kurt Lambeck, Patrick De Deckker, Paul Johnston, and L. Keith Fifield
2000 Timing of the Last Glacial Maximum from Observed Sea-Level Minima. *Nature* 406:713–716.

Manuscript received July 2, 2012.

Final revisions May 8, 2013.